



# PROJECT-BASED LEARNING

## PHASE-I EVALUATION

**AOL-Based Intelligent Tiered  
Memory Management System**

Team ID: T196

**Department of Computer Science & Engineering**  
Graphic Era (Deemed to be University), Dehradun  
Academic Session 2026–27

## Team & Mentor Details

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### Team Information

**Team ID:** T196

**Semester:** 5th

**Section:** ML1,ML2,DS1,C

**Domain:** *Operating Systems ,Database /*

### Team Members

1. *Garima Bhargava - 240121232*
2. *Nandini Bhatt - 24022559*
3. *Shaili Tomar - 240221680*
4. *Saumya Bisht – 240221898*

### Mentor

*Mr.Priyank Pandey*

**Interactions completed: 3** \_\_\_\_\_

# Problem Statement & Motivation

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## Problem Statement

*Existing memory systems mainly use access frequency to place data in fast memory. However, frequently accessed data is not always performance-critical. This project uses latency and memory parallelism for smarter data placement and reduced migration overhead.*

## Motivation

- ▶ *Existing systems mainly rely on access frequency (hotness).*
- ▶ *Important data stays in slow memory, hurting performance.*
- ▶ *Hotness ignores memory latency and parallel access.*

## Target Users / Stakeholders

- ▶ *Primary users: OS/kernel researchers.*
- ▶ *Secondary: cloud & CXL vendors.*
- ▶ *Benefit: better performance, less migration.*

# Objectives

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## Project Objectives

1. *Study existing memory tiering techniques and their frequency-based limitations.*
2. *Implement the Amortized Offcore Latency (AOL) score using latency and MLP data.*
3. *Build a placement method that prioritizes high-AOL data in fast memory.*
4. *Evaluate MemOpt against hotness-based methods using memory-intensive workloads.*
5. *Demonstrate reduced page migration and improved overall system performance.*

**Key Objective:** *Improve memory placement accuracy and reduce unnecessary page migrations using AOL-based decisions instead of access frequency.*

# Proposed Solution

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## Proposed Approach

1. *Collect memory access data via hardware counters (perf).*
2. *Compute latency and MLP to calculate AOL score.*
3. *Rank data by performance impact using AOL.*
4. *Place high-AOL data in fast memory, low-AOL in slower tiers.*
5. *Monitor performance and migration activity.*

*MemOpt replaces frequency-only hotness with a latency- and MLP-aware AOL score, prioritizing performance-critical data in fast memory while cutting wasteful migrations.*

## Key Features

- ▶ *Performance-aware placement, not just frequency-based.*
- ▶ *Amortized Offcore Latency (AOL) scoring.*
- ▶ *Reduced unnecessary page migrations.*
- ▶ *Real-time hardware profiling with perf.*
- ▶ *Benchmarked against traditional hotness methods.*

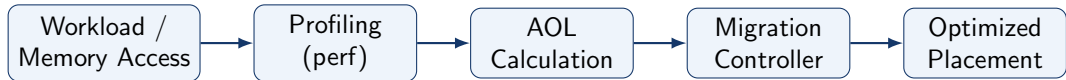
# Integration of PBL Course Concepts

| Course               | Concepts Applied                                                                                                                            | Project Module        |             |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------|
| Data Structures in C | Hashing for page-to-AOL score lookup, Trees/priority structures to rank pages by AOL, Linked Lists for tracking migration queues            | AOL Module            | Calculation |
| OOPs with C++        | Classes for memory tiers, profiling objects, and controller logic; Inheritance for tier-specific policies; STL containers for page metadata | Migration Controller  | Con-        |
| Operating Systems    | Processes, Memory Management, Paging, Memory Tiering, Scheduling of migration tasks                                                         | Memory Module         | Profiling   |
| DBMS                 | SQL, Transactions, Indexing, Buffer Pool Management, Query-aware page criticality (PostgreSQL)                                              | DBMS–MemOpt Interface |             |

**Requirement:** Demonstrate meaningful integration of both subjects in **one** project.

# System Architecture / Workflow

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## Major Components

- ▶ *Memory Profiling Module*
- ▶ *AOL Calculation Module*
- ▶ *Migration Controller*

## Data / Control Flow

- ▶ *Access patterns collected from workload.*
- ▶ *AOL score guides placement decisions.*
- ▶ *High-AOL data moved to fast memory.*

# Technology Stack & Methodology

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## Technology Stack

- ▶ Programming: C / C++ / Python
- ▶ Database: PostgreSQL
- ▶ Tools / Framework: Linux perf, Intel PMU / PEBS
- ▶ IDE: VS Code
- ▶ Version Control: Git / GitHub

## Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

C/C++ for low-level hardware/kernel access; PostgreSQL for DBMS-aware placement; perf & PMU/PEBS for latency/MLP profiling; Git/GitHub for team workflow.

# Initial Progress & Team Contribution

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## Progress Achieved

- ▶ Literature review of tiered memory systems (TPP, NUMA balancing) completed.
- ▶ Requirements for AOL-based placement identified.
- ▶ System architecture and module design prepared.
- ▶ Initial profiling module (perf integration) implemented.
- ▶ Test workload selection in progress.

## Individual Contribution

| Member  | Role      | Contribution     |
|---------|-----------|------------------|
| Garima  | Lead      | AOL design       |
| Nandini | Developer | Profiling module |
| Shaili  | Testing   | Benchmarking     |
| Saumya  | Docs      | Documentation    |

# Roadmap, Expected Outcomes & References

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## Roadmap

1. *Phase-I: Proposal & system design.*
2. *Phase-II: Profiling, AOL calculation & migration control.*
3. *Phase-III: Integration, testing & benchmarking.*

## Expected Outcomes

- ▶ *Working MemOpt prototype for tiered memory placement.*
- ▶ *Reduced page migration and CPU wait time.*
- ▶ *Applicability to real workloads.*
- ▶ *Future scalability to multi-tier systems.*

## References

1. *Liu et al., "Tiered Memory Beyond Hotness," OSDI 2025.*
2. *Linux Kernel Docs — Memory Management & Tiering.*
3. *Intel PMU & PEBS Documentation.*
4. *Linux perf Documentation.*

**Thank You**

Questions & Suggestions